

Nifty 100 Financial Intelligence Platform

Analyst Guide

Version 1.0 — Sprint 6

Table of Contents

- 1. Introduction & Platform Overview
- 2. Getting Started — Setup in Under 30 Minutes
- 3. Dashboard Guide — Home Screen
- 4. Dashboard Guide — Company Profile Screen
- 5. Dashboard Guide — Financial Screener
- 6. Dashboard Guide — Peer Comparison
- 7. Dashboard Guide — Trend, Sector & Capital Allocation Screens
- 8. Dashboard Guide — Annual Reports Screen
- 9. Generating PDF Tearsheets & Reports
- 10. Using the REST API — curl Examples
- 11. Troubleshooting Common Issues
- 12. Glossary of Key Terms

1. Introduction & Platform Overview

The Nifty 100 Financial Intelligence Platform is a self-contained analytics system covering all 92 Nifty 100 companies with data-availability history. It combines a cleaned SQLite database, a computed financial-ratio engine, an investment screener, peer comparison tools, a Streamlit dashboard, and a REST API — all built from 12 source datasets.

This guide is written for analysts who will use the dashboard and API day-to-day, not for the engineers who built the pipeline. Screenshots are described in words where a live screen capture isn't included, since screen layouts may be refreshed between releases.

2. Getting Started — Setup in Under 30 Minutes

Step 1: Clone or unzip the project

```
unzip nifty100_project.zip -d nifty100/
```

Step 2: Create a virtual environment

```
python -m venv venv  
source venv/Scripts/activate
```

Step 3: Install dependencies

```
pip install -r requirements.txt
```

Step 4: Build the database

```
python -m src.etl.db_loader
```

Step 5: Compute financial ratios

```
python -m src.analytics.compute_ratios
```

Step 6: Run the test suite

```
pytest tests/ -v
```

Step 7: Launch the dashboard

```
streamlit run src/dashboard/app.py
```

Step 8: Launch the API (separate terminal)

```
uvicorn src.api.main:app --port 8000
```

If setup succeeds, the dashboard opens automatically at <http://localhost:8501> and the API documentation is available at <http://localhost:8000/docs>.

3. Dashboard Guide — Home Screen

The Home screen is the landing page, showing 6 summary KPI tiles (average ROE, median P/E, median D/E, total companies, median revenue CAGR, and debt-free company count), a sector-breakdown donut chart, and a top-5 companies table by composite quality score.

Use the year selector in the sidebar to view metrics as of an earlier year.

4. Dashboard Guide — Company Profile Screen

Search any of the 92 companies by name or ticker. The profile shows a company card (sector, description), 6 KPI tiles, a 10-year revenue and profit chart, an ROE/ROCE trend line, and pros/cons as coloured badges.

If a ticker cannot be found, a friendly message appears rather than an error.

5. Dashboard Guide — Financial Screener

Use the 10 sidebar sliders to set your own thresholds, or click one of the 6 preset buttons (Quality, Value, Growth, Dividend, Debt-Free, Turnaround) to auto-fill sensible starting values.

The results table updates live as you move a slider. Use the CSV download button to export the current result set.

Note: the Debt-to-Equity filter automatically excludes Financials-sector companies, since high leverage is structurally normal for banks and NBFCs.

6. Dashboard Guide — Peer Comparison

Select a peer group from the dropdown to see a radar chart comparing a chosen company against its peer group average across 8 metrics, plus a side-by-side table with the benchmark company highlighted in gold.

7. Dashboard Guide — Trend, Sector & Capital Allocation

Trend Analysis overlays up to 3 metrics over a 10-year history with year-over-year change shown on hover.

Sector Analysis shows a revenue-vs-ROE bubble chart and sector median bars.

Capital Allocation Map is a treemap grouping all companies into 8 capital-allocation patterns (Reinvestor, Distress Signal, etc.) — click a pattern to see the company list.

8. Dashboard Guide — Annual Reports Screen

Search a company to see its available annual report links by year, with a red 'Report unavailable' badge if a link no longer resolves.

9. Generating PDF Tearsheets & Reports

Company tearsheets, sector reports, and the portfolio summary are pre-generated batch outputs, not created live from the dashboard. To regenerate them after a data refresh:

```
python -m src.reports.batch_tearsheets
```

```
python -m src.reports.sector_report
```

```
python -m src.reports.portfolio_summary
```

Tearsheets land in reports/tearsheets/, sector reports in reports/sector/, and the portfolio summary in reports/portfolio/. A company with fewer than 3 years of history is skipped and logged to output/skipped_tearsheets.csv rather than causing an error.

10. Using the REST API — curl Examples

Health check

```
curl http://localhost:8000/api/v1/health
```

List all companies

```
curl http://localhost:8000/api/v1/companies
```

Filter by sector

```
curl "http://localhost:8000/api/v1/companies?sector=IT"
```

Company profile

```
curl http://localhost:8000/api/v1/companies/TCS
```

Company ratios history

```
curl http://localhost:8000/api/v1/companies/TCS/ratios
```

Run the screener

```
curl "http://localhost:8000/api/v1/screener?min_roe=15&max_de=1"
```

Sector summary

```
curl http://localhost:8000/api/v1/sectors
```

Peer group data

```
curl "http://localhost:8000/api/v1/peers/IT%20Services"
```

Download a tearsheet

```
curl -o TCS.pdf http://localhost:8000/api/v1/companies/TCS/tearsheet
```

The full interactive API documentation (OpenAPI / Swagger UI) is available at <http://localhost:8000/docs> whenever the API server is running, and lets you try every endpoint directly from the browser.

11. Troubleshooting Common Issues

Issue: Dashboard shows 'No data available'

Fix: Confirm the database has been built (`python -m src.etl.db_loader`) and the `financial_ratios` table is populated (`python -m src.analytics.compute_ratios`).

Issue: Port already in use

Fix: Another process is likely using port 8501 or 8000. Stop it, or run with a different port: `streamlit run src/dashboard/app.py --server.port 8502`

Issue: 'Ticker not found' for a company you expect to exist

Fix: Tickers are case-sensitive and normalised to upper-case, no spaces. Search by company name instead if unsure of the exact ticker.

Issue: Tearsheet download returns 404

Fix: The PDF hasn't been generated yet for that company, or it was skipped for having fewer than 3 years of history. Check `output/skipped_tearsheets.csv`.

Issue: Screener returns 0 companies

Fix: Thresholds may be too strict for the current data. Try the preset buttons first, then loosen one filter at a time.

Issue: API returns HTTP 500

Fix: Check the `DB_PATH` environment variable points at the correct database file, and that the database has been built.

12. Glossary of Key Terms

ROE	Return on Equity — net profit divided by shareholder equity.
ROCE	Return on Capital Employed — a broader profitability measure.
D/E	Debt-to-Equity — a leverage ratio; 0 means debt-free.
CAGR	Compound Annual Growth Rate — annualised growth over a period.
FCF	Free Cash Flow — cash generated after operating and investing activity.
ICR	Interest Coverage Ratio — ability to service debt from operating profit.
Peer Group	A set of comparable companies used for relative ranking.
Composite Score	A 0-100 weighted quality score combining profitability, cash quality, growth, and leverage.